Date – 09/03/2026

EXPERIMENT 4

AIM OF THE EXPERIMENT –

To build a system in Simulink that amplifies a sine wave by a factor of M and also test the linearity of system using SIMULINK.

OBJECTIVE -:

1. Test the Linearity of the system y(t) = 2x(t) + 1 using Simulink.
2. Test the Linearity of the system y(t) = x^2(t) using Simulink.
3. Test the Linearity of the system y(t) = x(t-2) using Simulink.

SOFTWARE REQUIRED –

SIMULINK R2025b

THEORY -

Simulink is a graphical simulation environment provided in MATLAB that is used to model, simulate, and analyze dynamic systems. It allows users to build systems using block diagrams instead of writing lengthy codes. Different blocks such as sources, mathematical operations, signal routing, and sinks can be connected together to represent a complete system. Simulink is widely used in signal processing, communication systems, and control systems to study the behavior of signals and systems through simulation.

A **Subsystem** in Simulink is a block that contains a group of other blocks combined into a single unit. It helps in organizing complex models by grouping related components together. Subsystems improve readability, reduce clutter in the main model, and allow the designer to reuse the same block structure multiple times. Inputs and outputs of a subsystem are defined using Inport and Outport blocks, making it behave like a single functional block within the larger Simulink model.

A **linear system** in signal processing is a system that satisfies the principle of superposition. This means the output of the system changes proportionally with the input. If the properties of linearity are not satisfied, the system is considered a non-linear system.

There are two common methods used to test the linearity of a system.

Superposition Principle

The Superposition Principle states that the response of a system to a weighted sum of inputs is equal to the weighted sum of the responses of each individual input.

Mathematically, it is expressed as:

$$\boldsymbol{T}[\boldsymbol{a}\boldsymbol{x}_1(\boldsymbol{t}) + \boldsymbol{b}\boldsymbol{x}_2(\boldsymbol{t})] = \boldsymbol{a}\boldsymbol{T}[\boldsymbol{x}_1(\boldsymbol{t})] + \boldsymbol{b}\boldsymbol{T}[\boldsymbol{x}_2(\boldsymbol{t})]$$

where

- Trepresents the system or transformation,
- $x_1(t)$and $x_2(t)$are input signals,
- $a$and $b$are constants,
- $T[x_1(t)]$and $T[x_2(t)]$are the outputs corresponding to the inputs.

This formula means that if two signals $x_1(t)$and $x_2(t)$are applied to a system separately to

produce outputs $T[x_1(t)]$and $T[x_2(t)]$, then applying the combined input $ax_1(t) + bx_2(t)$should produce an output equal to $aT[x_1(t)] + bT[x_2(t)]$. If this condition is satisfied, the system obeys the superposition principle and is considered linear.

Law of Additivity and Homogeneity

Another way to verify linearity is by checking two properties: additivity and homogeneity (scaling).

Law of Additivity:

If a system produces an output $y_1(t)$for input $x_1(t)$and an output $y_2(t)$for input $x_2(t)$, then for the combined input $x_1(t) + x_2(t)$, the output y_3 should be equal to $y_1(t) + y_2(t)$.

Law of Homogeneity (Scaling):

If an input $x(t)$produces an output $y(t)$, then a scaled input $a\,x(t)$should produce a scaled output $a\,y(t)$, where $a$is a constant.

A system is linear only if both additivity and homogeneity are satisfied simultaneously.

Theoretical Calculation -:

1. **y(t) = 2x(t) + 1**
   - Test for Additivity

Let the inputs be $x_1(t)$and $x_2(t)$.

For input $x_1(t)$:

$$y_1(t) = 2x_1(t) + 1$$

For input $x_2(t)$:

$$y_2(t) = 2x_2(t) + 1$$

For the combined input $x_1(t) + x_2(t)$:

$$y_3(t) = 2(x_1(t) + x_2(t)) + 1$$
$$y_3(t) = 2x_1(t) + 2x_2(t) + 1$$

Now,

$$y_1(t) + y_2(t) = (2x_1(t) + 1) + (2x_2(t) + 1)$$
$$y_1(t) + y_2(t) = 2x_1(t) + 2x_2(t) + 2$$

Since

$$y_3(t) \neq y_1(t) + y_2(t)$$

the additivity property is not satisfied.

- Test for Homogeneity

Let the input be scaled by a constant a.

For input x(t):

$$y(t) = 2x(t) + 1$$

For input ax(t):

$$T[ax(t)] = 2(ax(t)) + 1$$
$$T[ax(t)] = 2a\,x(t) + 1$$

But

$$aT[x(t)] = a(2x(t) + 1)$$
$$aT[x(t)] = 2a\,x(t) + a$$

Since

$$T[ax(t)] \neq aT[x(t)]$$

the homogeneity property is also not satisfied.

Since the system does not satisfy additivity and homogeneity, it violates the superposition principle. Therefore, the system

$$y(t) = 2x(t) + 1$$

is a non-linear system.

2. $y(t) = x^2(t)$

Let the system be

$$T[x(t)] = x^2(t)$$

Take two input signals $x_1(t)$and $x_2(t)$.

For input $x_1(t)$:

$$T[x_1(t)] = x_1^2(t)$$

For input $x_2(t)$:

$$T[x_2(t)] = x_2^2(t)$$

For the combined input $ax_1(t) + bx_2(t)$:

$$T[ax_1(t) + bx_2(t)] = (ax_1(t) + bx_2(t))^2$$

Expanding,

$$T[ax_1(t) + bx_2(t)] = a^2x_1^2(t) + b^2x_2^2(t) + 2ab\, x_1(t)x_2(t)$$

Now,

$$aT[x_1(t)] + bT[x_2(t)] = ax_1^2(t) + bx_2^2(t)$$

Since

$$(ax_1(t) + bx_2(t))^2 \neq ax_1^2(t) + bx_2^2(t)$$

the superposition condition is not satisfied.

Because the system does not satisfy the superposition principle, the system

$$y(t) = x^2(t)$$

is a non-linear system.

3. $y(t) = x(t-2)$

Let the system be

$$T[x(t)] = x(t-2)$$

Take two input signals $x_1(t)$and $x_2(t)$.

For input $x_1(t)$:

$$T[x_1(t)] = x_1(t-2)$$

For input $x_2(t)$:

$$T[x_2(t)] = x_2(t-2)$$

For the combined input $ax_1(t) + bx_2(t)$:

$$T[ax_1(t) + bx_2(t)] = ax_1(t-2) + bx_2(t-2)$$

Now,

$$aT[x_1(t)] + bT[x_2(t)] = ax_1(t-2) + bx_2(t-2)$$

Since

$$T[ax_1(t) + bx_2(t)] = aT[x_1(t)] + bT[x_2(t)]$$

the superposition principle is satisfied.

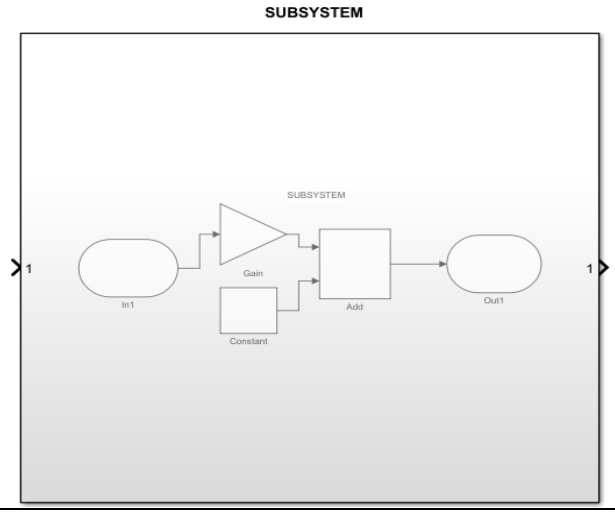
Since the system satisfies the superposition principle, the system

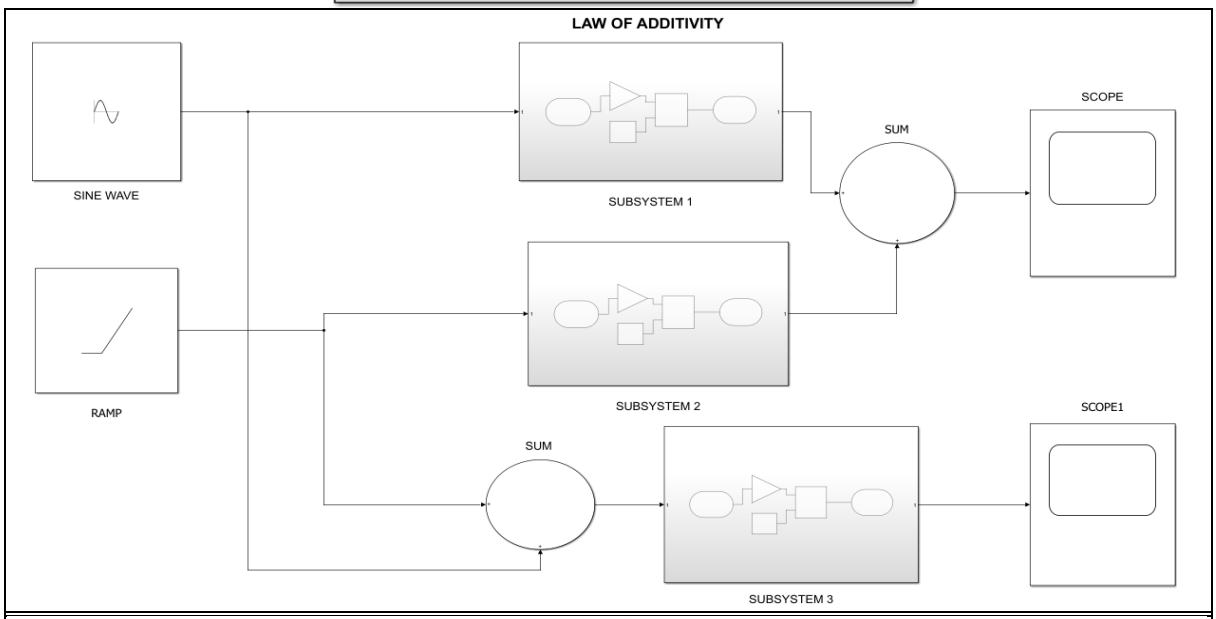
$$y(t) = x(t-2)$$

is a linear system.

Block Diagram -:

1. OBJECTIVE 1 (Test the Linearity of the system y(t) = 2x(t) + 1 using Simulink.)





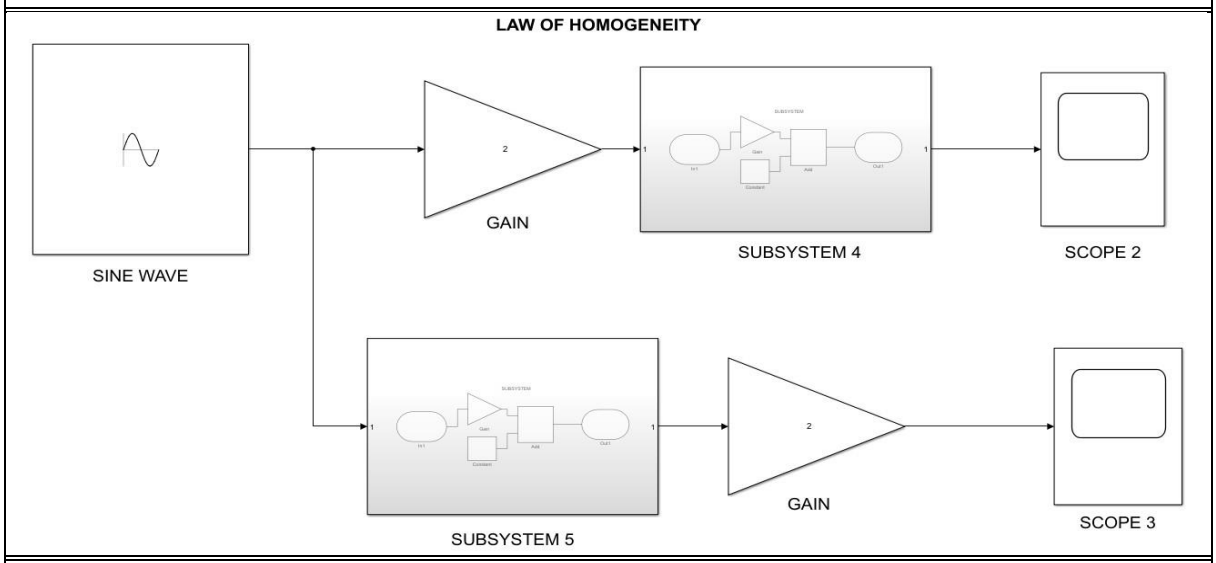


Fig 4.1 Block Diagram for linearity test of y(t)=2x(t) + 1 using Simulink

RESULT

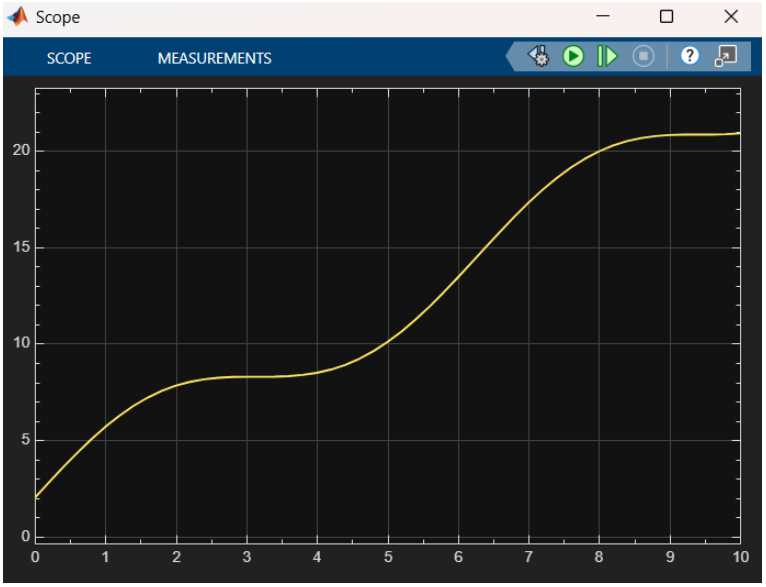


Fig 4.2 Output of $y_1(t) + y_2(t)$

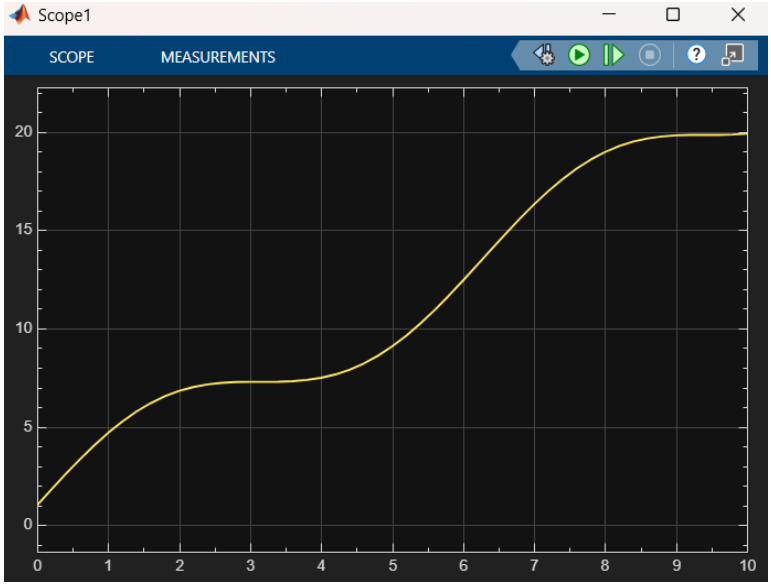


Fig 4.3 Output of $y_3(t)$

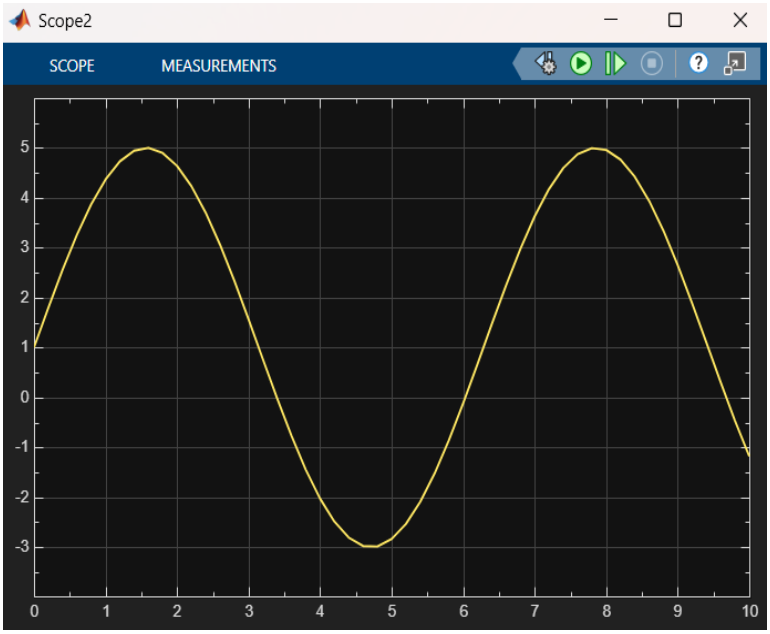


Fig 4.4 Output of T[ax(t)]

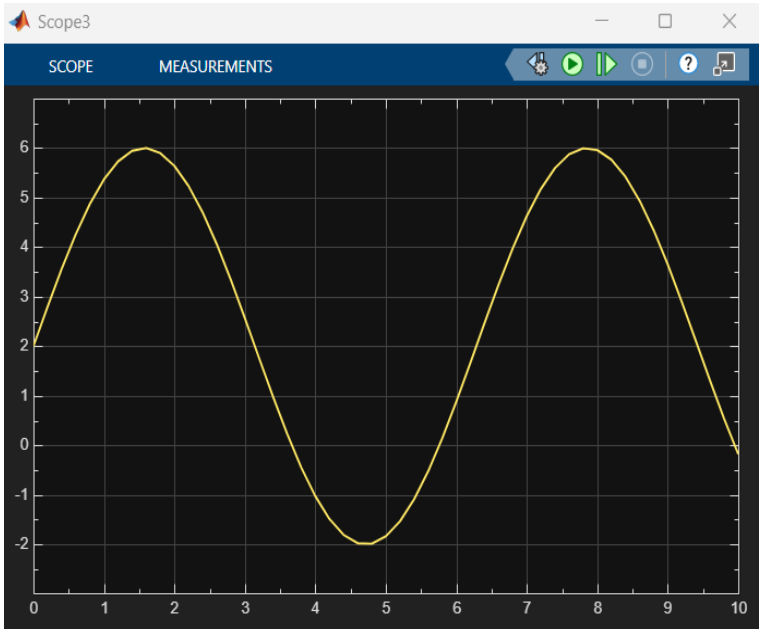
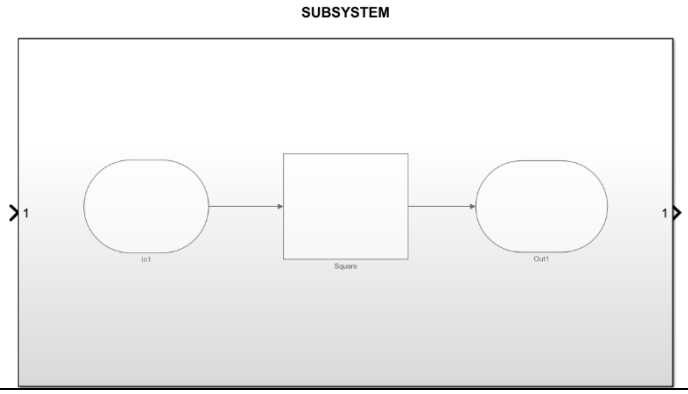


Fig 4.5 Output of aT[x(t)]

2. OBJECTIVE 2 (Test the Linearity of the system $y(t) = x^2(t)$ using Simulink.)



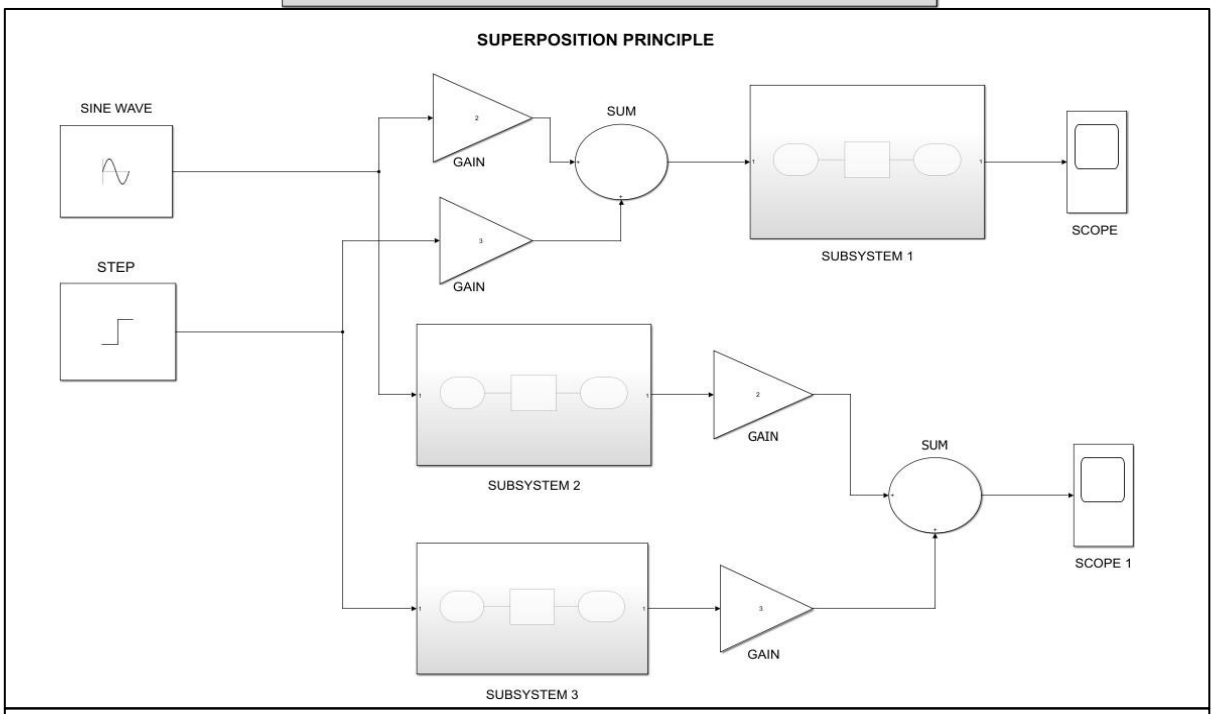


Fig 4.6 Block Diagram for linearity test of $y(t)= x^2(t)$ using Simulink

RESULT

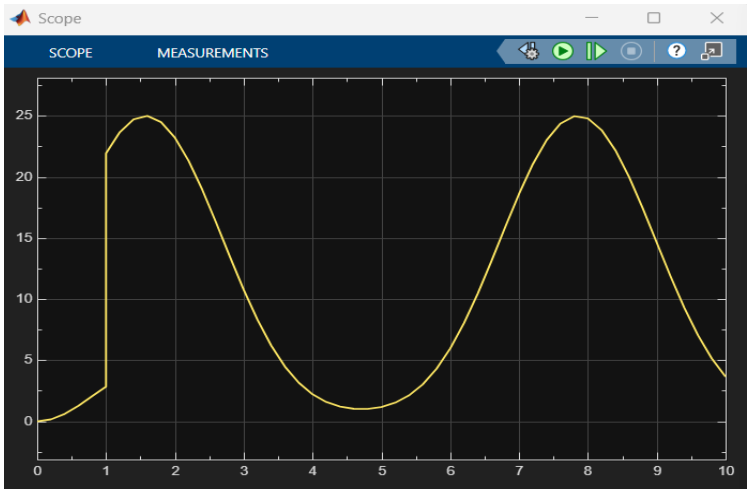


Fig 4.7 Output of $T[ax_1(t) + bx_2(t)]$

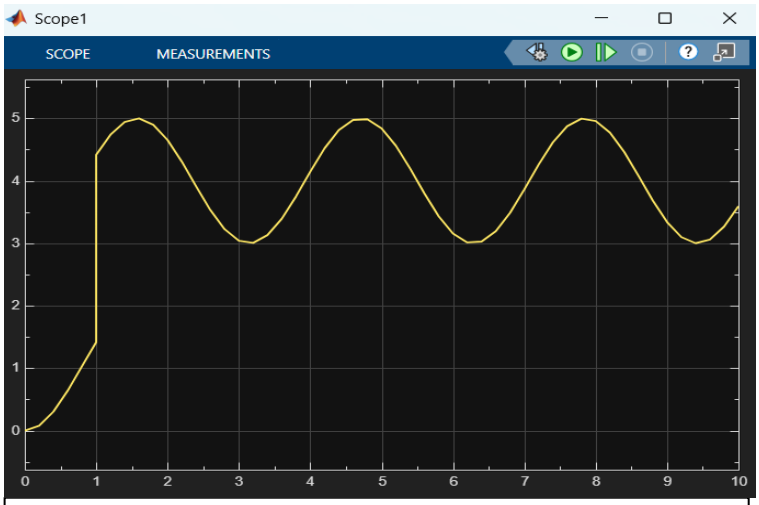
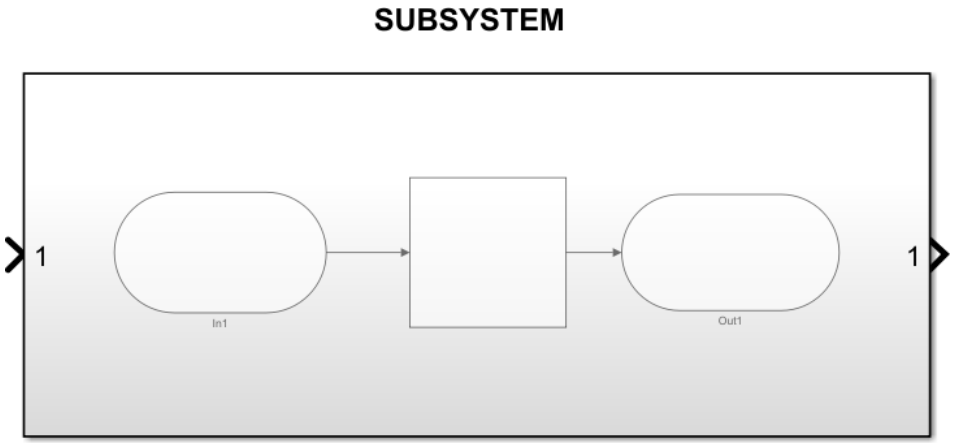


Fig 4.8 Output of $aT[x_1(t)] + bT[x_2(t)]$

3. OBJECTIVE 3 (Test the Linearity of the system y(t) = x(t-2) using Simulink.)



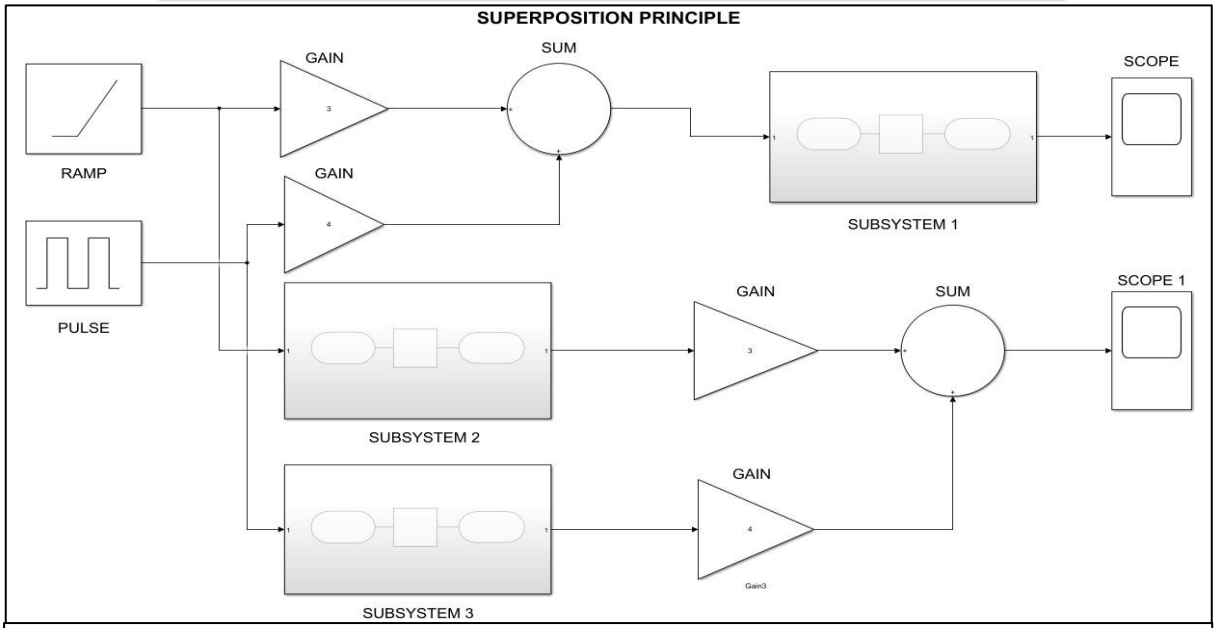


Fig 4.9 Block Diagram for linearity test of y(t)= x(t-2) using Simulink

RESULT

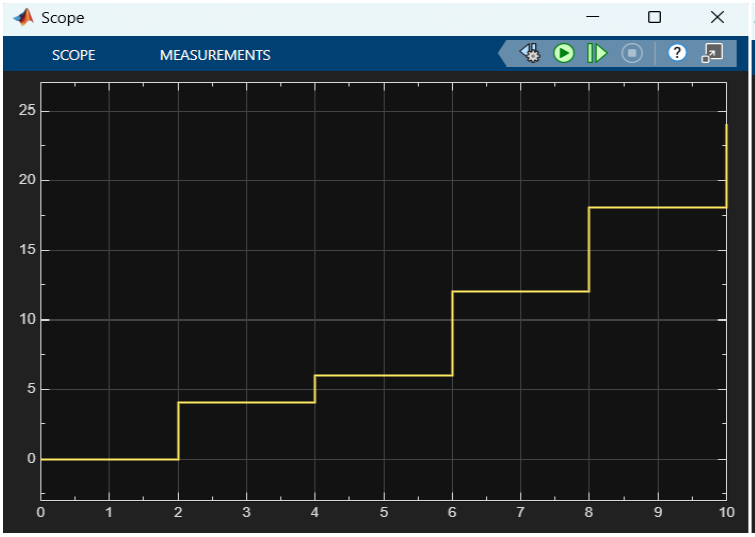


Fig 4.10 Output of $T[ax_1(t) + bx_2(t)]$

Fig 4.8 Output of $aT[x_1(t)] + bT[x_2(t)]$

CONCLUSION -

In this experiment, different systems were implemented in MATLAB Simulink to study the amplification of a sine wave and to test the linearity of given systems using the principle of superposition. The system $y(t) = 2x(t) + 1$was found to be non-linear because of the presence of a constant term. The system $y(t) = x^2(t)$was also non-linear since it does not satisfy the superposition property. However, the system $y(t) = x(t-2)$satisfied the linearity condition and behaved as a linear time-delay system. Thus, the experiment successfully demonstrated how Simulink can be used to analyse and verify the linearity properties of different systems.

Submitted By

Signature of Faculty